



Вселенная

Статистика на Python: 100 решенных задач по анализу данных

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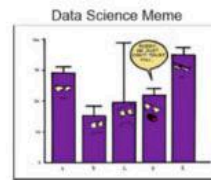
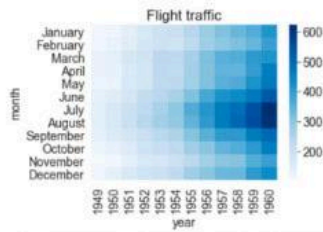
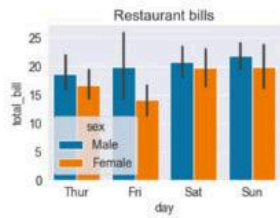
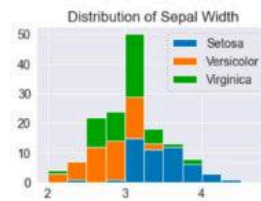
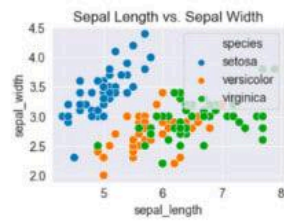
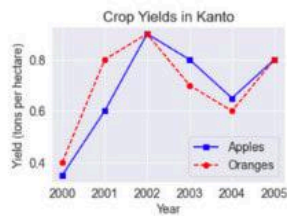
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Статистика на Python: 100 решенных задач по анализу данных

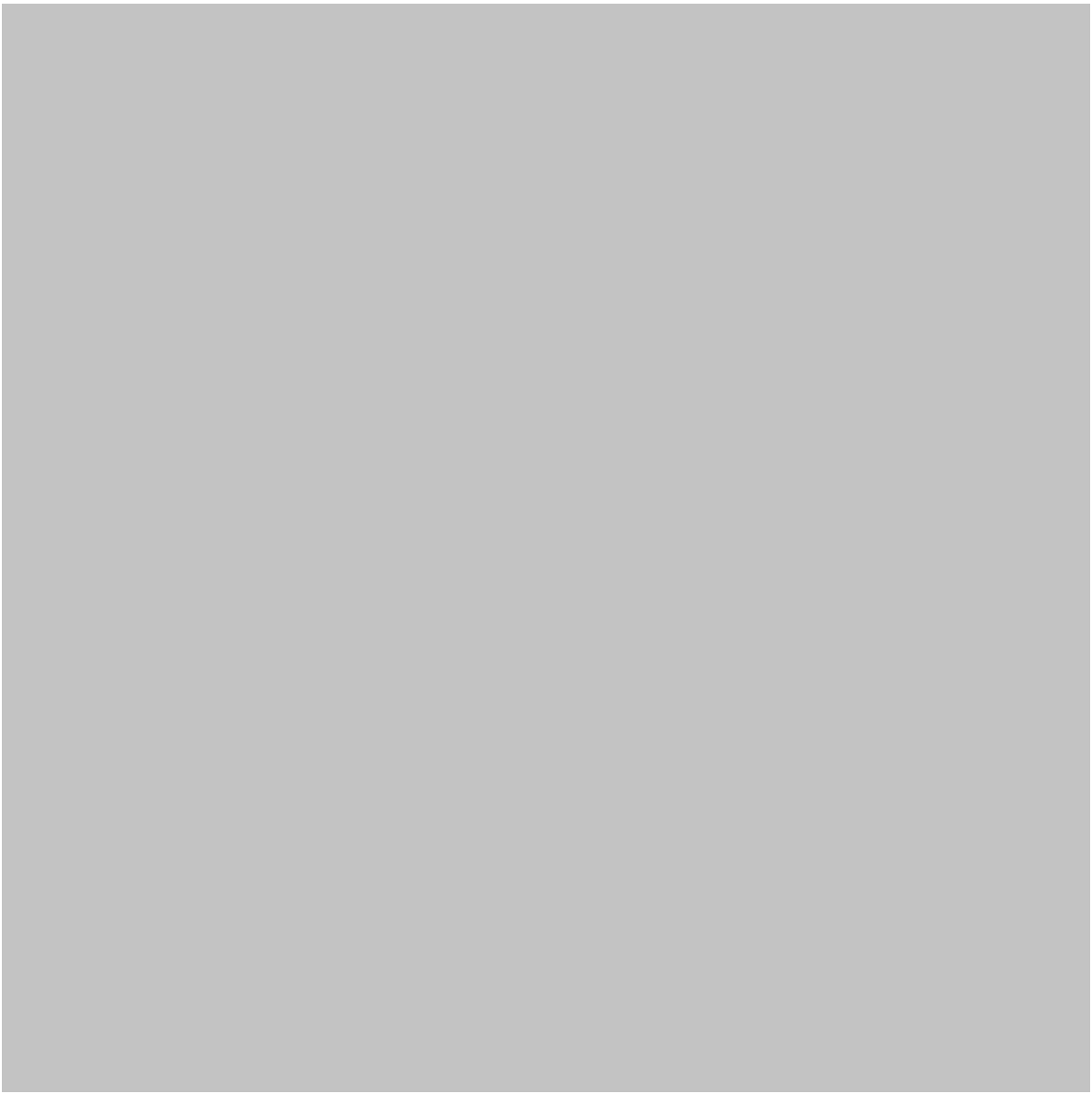
Введение

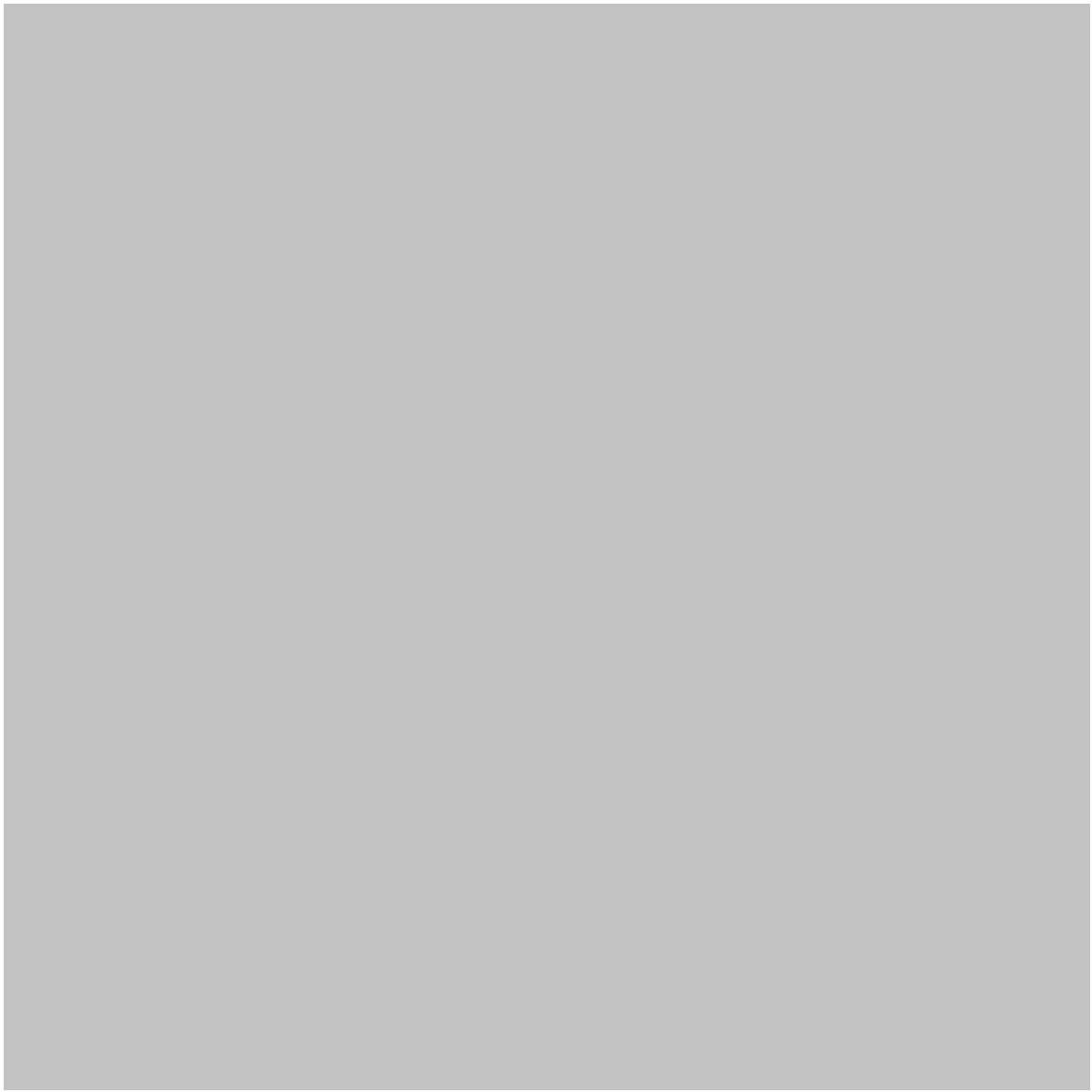
Статистика — одна из важнейших основ современного анализа данных. Независимо от того, изучаете ли вы инженерное дело, финансы, бизнес, информатику, искусственный интеллект или занимаетесь научными исследованиями, статистическое мышление помогает преобразовывать необработанные наблюдения в полезные выводы. 📊

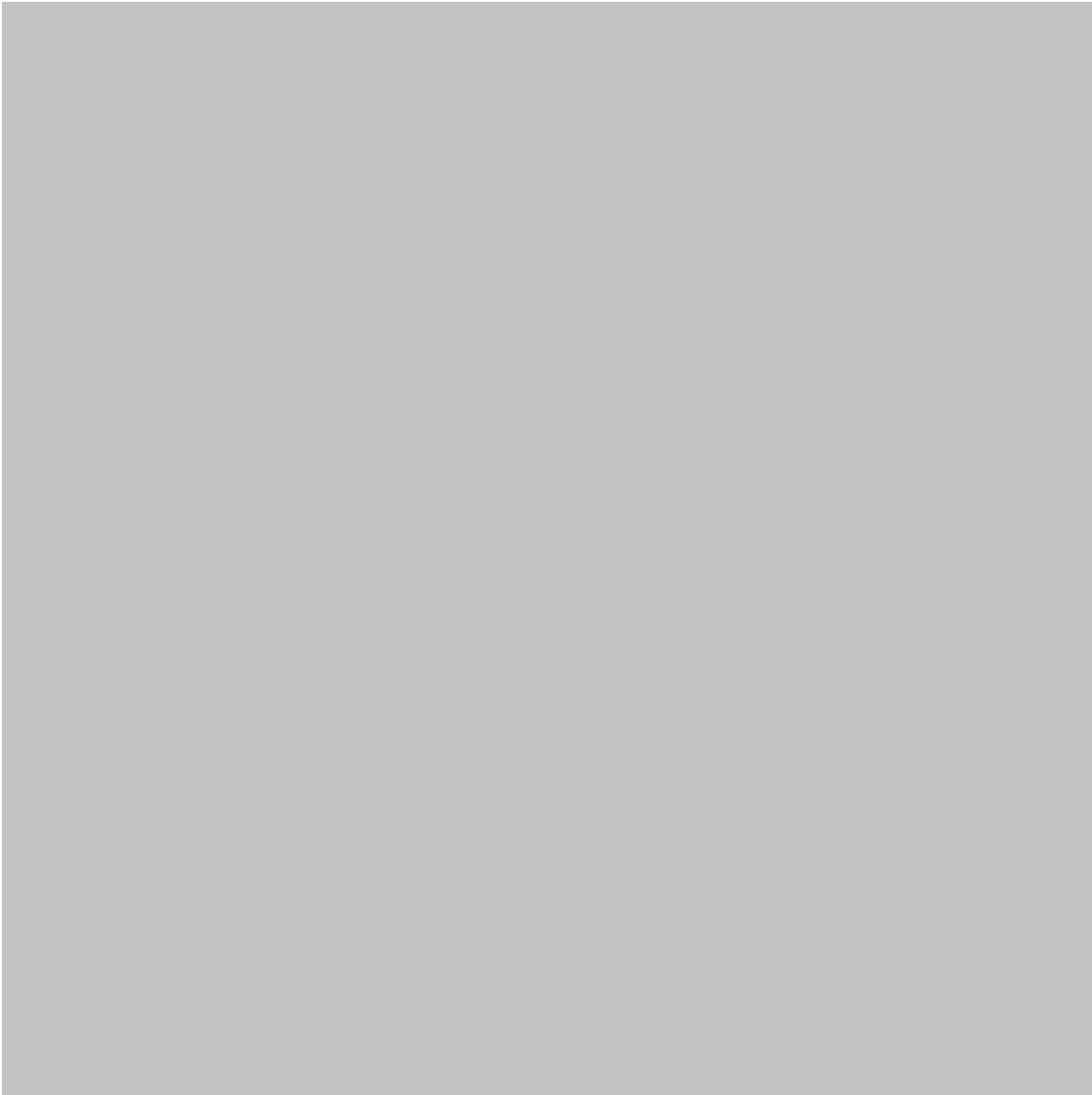
Python делает статистический анализ особенно доступным, поскольку сочетает в себе понятный язык программирования с мощными библиотеками, такими как NumPy, pandas, SciPy, Matplotlib и Statsmodels.



DATA ANALYSIS USING PYTHON







В этом практическом руководстве представлены **100 решенных статистических задач для анализа данных с помощью Python**. Упражнения охватывают все аспекты — от базовой описательной статистики до теории вероятностей, распределений, выборки, проверки гипотез, корреляции, регрессии и аналитической интерпретации реальных данных.

Цель состоит не только в том, чтобы подсчитать статистические показатели. Профессиональный аналитик также должен понимать, **что означают результаты, насколько надежны данные и как полученные выводы могут повлиять на принятие решения**. 🔍

Теоретическая база

Статистику можно условно разделить на две основные области: **описательную статистику и статистику, основанную на выводах**.

Описательная статистика

Описательная статистика обобщает информацию, уже представленную в наборе данных.

К распространенным методам относятся:

- Среднее значение
- Медиана
- Мода
- Минимальное и максимальное значения
- Размах
- Дисперсия
- Стандартное отклонение
- Квартили
- Процентили
- Частотные распределения
- Визуализация данных

Например, инженер может собрать данные об измерениях 500 изготовленных деталей. Вместо того чтобы проверять каждое измерение по отдельности, можно использовать описательную статистику, чтобы выявить типичные значения, отклонения, необычные наблюдения и общее распределение.

Статистический вывод

Статистический вывод — это следующий шаг. Он использует выборку для того, чтобы сделать выводы о более крупной совокупности.

К типичным областям применения относятся:

- Доверительные интервалы
- Проверка гипотез
- Статистическая значимость
- Корреляционный анализ
- Регрессия
- Дисперсионный анализ
- Вероятностные модели

Это различие очень важно: **описательная статистика описывает наблюдаемые данные, а индуктивная статистика помогает делать выводы, выходящие за рамки наблюдаемой выборки.**

Статистическая экосистема Python

Python предоставляет несколько дополнительных инструментов:

Library	Main purpose
NumPy	Numerical calculations
pandas	Data manipulation
SciPy	Scientific and statistical analysis
Matplotlib	Visualization
Seaborn	Statistical visualization
Statsmodels	Statistical models and inference
scikit-learn	Predictive modeling

Definition

Statistics with Python is the application of statistical concepts and analytical methods through Python programming tools to collect, organize, summarize, visualize, analyze, and interpret data.

A typical workflow looks like this:

Data → Cleaning → Exploration → Statistics → Visualization → Interpretation → Decision

The important point is that Python does not replace statistical reasoning. It automates calculations and provides analytical tools, but the analyst remains responsible for selecting appropriate methods and interpreting the results correctly.

Step-by-Step Explanation

A reliable statistical analysis normally follows a structured workflow.

Step 1: Import the Data

Data may originate from:

- CSV files

- Excel spreadsheets
- Databases
- Sensors
- Surveys
- APIs
- Laboratory experiments
- Website analytics

Pandas is commonly used to load structured datasets.

Step 2: Inspect the Dataset

Before performing statistical calculations, inspect:

- Number of rows
- Number of columns
- Column names
- Data types
- Missing observations
- Duplicate records
- Unusual values

⚠️ Never begin statistical modeling before understanding the structure of the data.

Step 3: Clean the Data

Cleaning can include:

- Removing duplicates
- Correcting data types
- Handling missing values
- Detecting impossible observations
- Standardizing categories
- Identifying outliers

Step 4: Calculate Descriptive Statistics

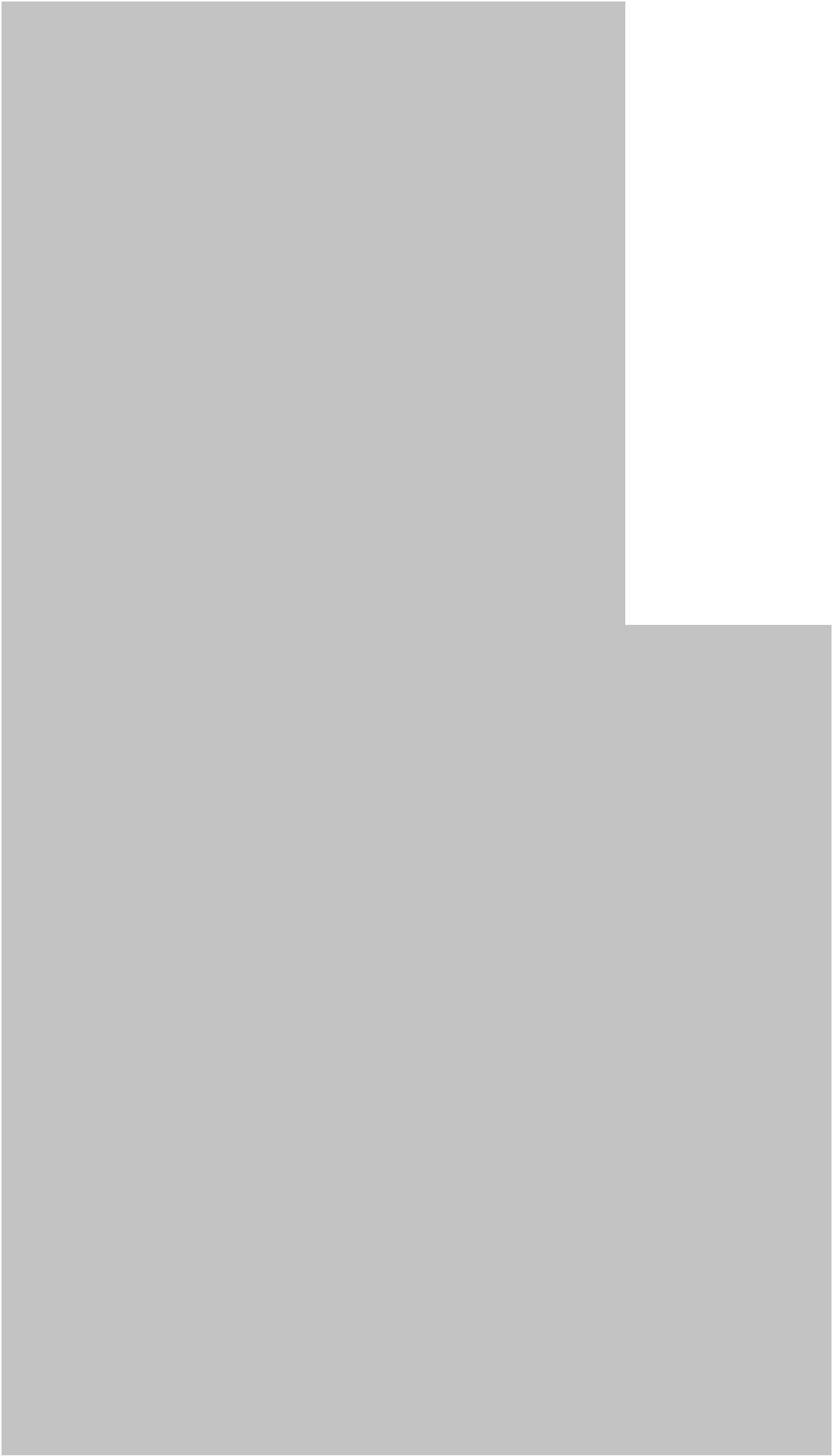
Python can quickly summarize numerical columns and reveal patterns that may not be obvious from the raw dataset.

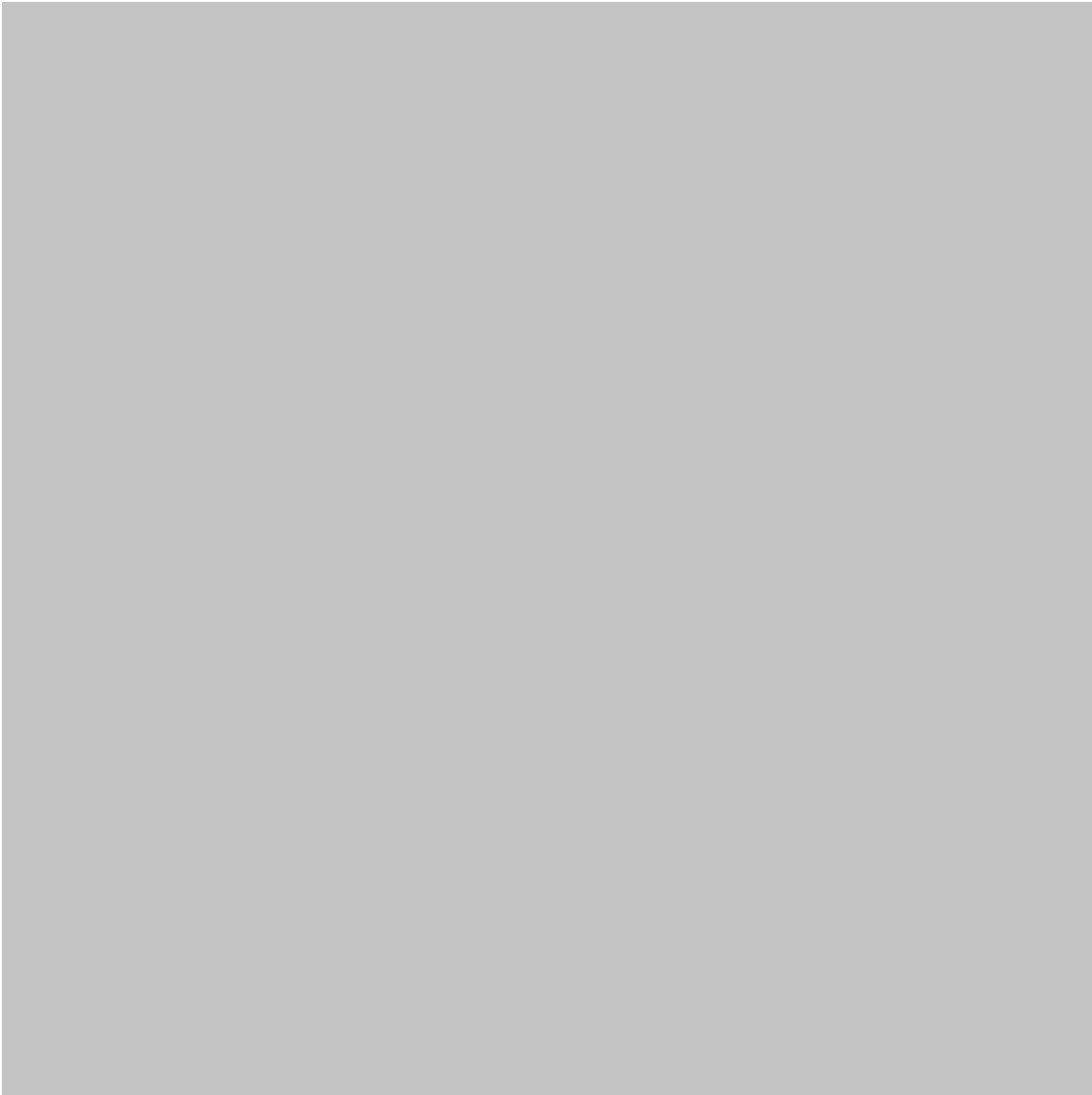
Step 5: Visualize the Data

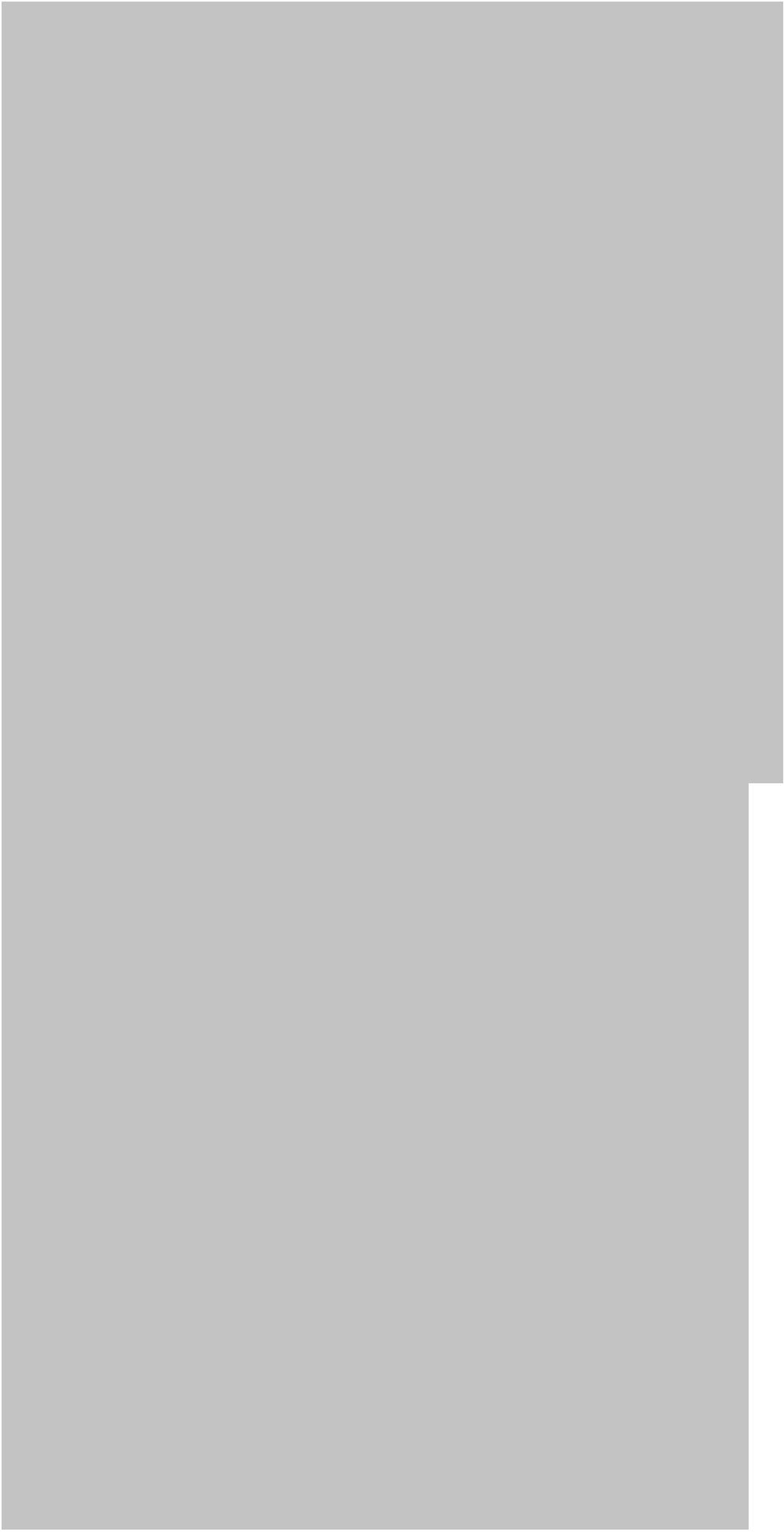
Charts help identify:

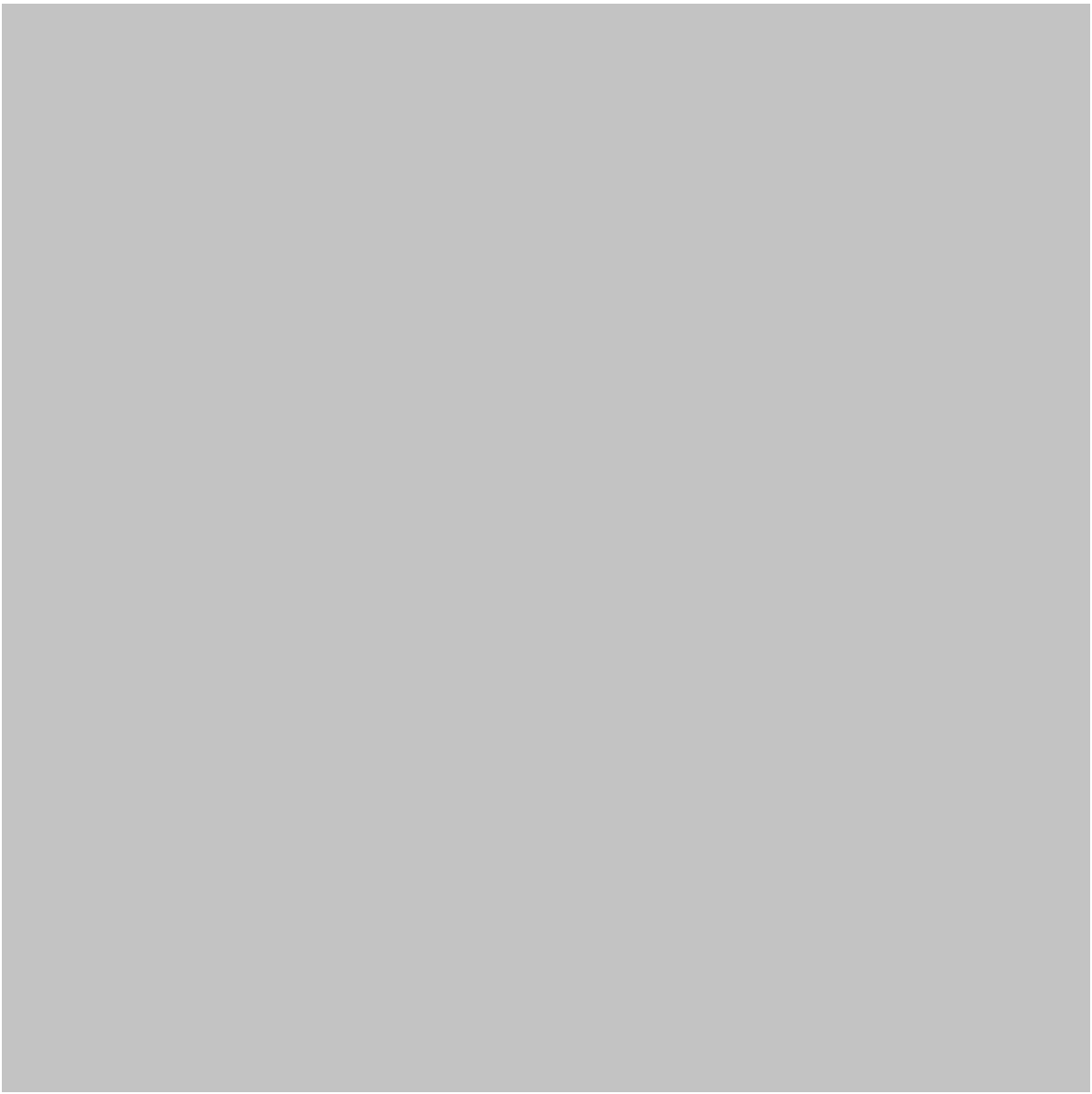
- Skewness

- Clusters
- Trends
- Outliers
- Relationships
- Distribution shapes











Step 6: Select an Appropriate Statistical Method

The method depends on the question and data.

For example:

- Comparing groups → t-test or ANOVA
- Examining relationships → correlation
- Predicting a numerical variable → regression
- Studying categorical relationships → chi-square analysis
- Examining distributions → distribution analysis

Step 7: Interpret the Results

The final result should answer a practical question.

A statistical output by itself is not a conclusion.

100 Solved Exercises

The following exercises form a compact practice sequence. The solutions describe the expected analytical result or Python approach without relying on lengthy mathematical notation.

Exercises 1–10: Data Fundamentals

1. Load a CSV dataset

Solution: Use pandas to read the CSV into a DataFrame, then inspect its first records.

2. Count observations

Solution: Examine the DataFrame dimensions to determine the number of records.

3. Identify numerical columns

Solution: Use pandas data-type information to separate numerical variables from categorical variables.

4. Find missing values

Solution: Apply pandas missing-value detection and summarize missing observations by column.

5. Remove duplicate records

Solution: Detect duplicated rows and retain one copy of each unique observation.

6. Examine column names

Solution: Display the DataFrame column index and verify that names are meaningful.

7. Generate a statistical summary

Solution: Use pandas descriptive-statistics functionality to obtain central tendency and variation information.

8. Sort observations

Solution: Sort the selected variable in ascending or descending order and inspect extreme observations.

9. Filter observations

Solution: Apply a Boolean condition to select records satisfying a specified analytical criterion.

10. Select a random sample

Solution: Use pandas sampling functionality to select representative observations for exploratory analysis.

Exercises 11–20: Central Tendency

11. Find the mean

Solution: Use the pandas mean method on the selected numerical column.

12. Find the median

Solution: Apply the median method and compare it with the mean.

13. Find the mode

Solution: Use the mode method and inspect whether one or multiple values dominate.

14. Compare mean and median

Solution: A large difference can indicate skewness or extreme observations.

15. Find the minimum

Solution: Use the minimum function to identify the smallest recorded observation.

16. Find the maximum

Solution: Use the maximum function to identify the largest observation.

17. Determine the range

Solution: Compare the minimum and maximum values to understand the overall spread.

18. Calculate quartiles

Solution: Use pandas quantile functionality to identify distribution boundaries.

19. Find the interquartile range

Solution: Compare the upper and lower quartiles to measure the spread of the central portion of the data.

20. Identify the most representative statistic

Solution: When data contains strong outliers, the median can provide a more robust description of the center.

Exercises 21–30: Variability

21. Calculate variance

Solution: Use pandas or NumPy variance functionality while checking whether sample or population interpretation is appropriate.

22. Calculate standard deviation

Solution: Apply the standard deviation method and interpret it as a measure of data dispersion.

23. Compare two datasets

Solution: Compare their standard deviations to determine which dataset has greater variability.

24. Detect high variability

Solution: Examine dispersion relative to the typical values rather than considering the standard deviation alone.

25. Find extreme observations

Solution: Sort observations or use quartile-based methods to investigate unusually large or small values.

26. Calculate percentiles

Solution: Use pandas quantile functionality to locate observations at selected distribution positions.

27. Create a box plot

Solution: Use Matplotlib or Seaborn to visualize median, quartiles, and potential outliers.

28. Analyze data consistency

Solution: A narrow distribution indicates greater consistency, while a wide distribution indicates greater variation.

29. Compare engineering measurements

Solution: Compare both central tendency and dispersion rather than relying on

averages alone.

30. Identify the most stable process

Solution: The process with lower meaningful variability is generally more consistent, provided the measurement scales are comparable.

Exercises 31–40: Probability

31. Count successful outcomes

Solution: Filter observations satisfying the required condition and count them.

32. Calculate observed frequency

Solution: Divide the number of qualifying observations by the total number of observations.

33. Estimate event probability

Solution: Use observed frequencies when estimating probability from historical data.

34. Analyze repeated events

Solution: Group observations according to event outcomes and compare their frequencies.

35. Create a probability table

Solution: Use pandas value counts and normalize the results to obtain relative frequencies.

36. Analyze categorical outcomes

Solution: Count each category and visualize the distribution with a bar chart.

37. Compare observed and expected patterns

Solution: Examine whether observed frequencies approximately follow the expected pattern.

38. Simulate random events

Solution: NumPy can generate repeated random outcomes and help demonstrate probabilistic behavior.

39. Increase simulation size

Solution: Larger simulations generally provide more stable estimates of expected behavior.

40. Visualize simulated probability

Solution: Plot the simulated outcomes and inspect how the distribution changes as observations increase.

Exercises 41–50: Distributions

41. Create a histogram

Solution: Use Matplotlib or Seaborn to display the frequency distribution of a numerical variable.

42. Identify skewness visually

Solution: Examine whether observations are concentrated toward one side with a longer tail.

43. Examine a normal-looking distribution

Solution: Use a histogram and additional diagnostic plots rather than assuming normality from appearance alone.

44. Compare distributions

Solution: Plot two datasets using consistent scales and compare their centers and spreads.

45. Identify outliers visually

Solution: Use box plots and histograms to detect unusual observations.

46. Examine distribution symmetry

Solution: Compare the position of the center with the shape of the tails.

47. Standardize observations

Solution: Use appropriate NumPy or SciPy functions to transform variables for comparative analysis.

48. Analyze sampling behavior

Solution: Repeatedly draw samples and examine how sample statistics vary.

49. Study sampling distributions

Solution: Simulation demonstrates how estimates behave across repeated samples.

50. Visualize distribution changes

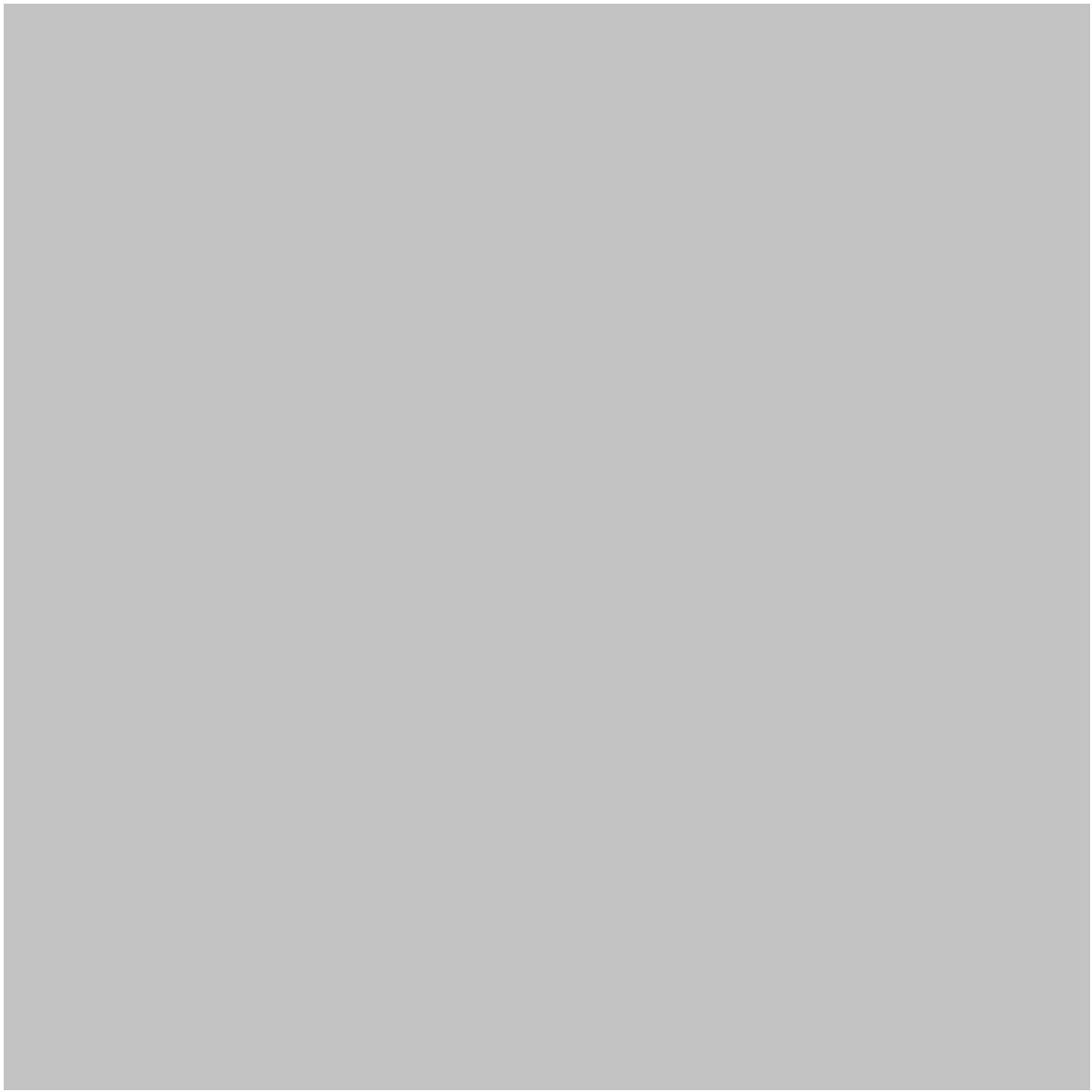
Solution: Generate multiple samples and plot their statistical summaries to observe sampling variability.

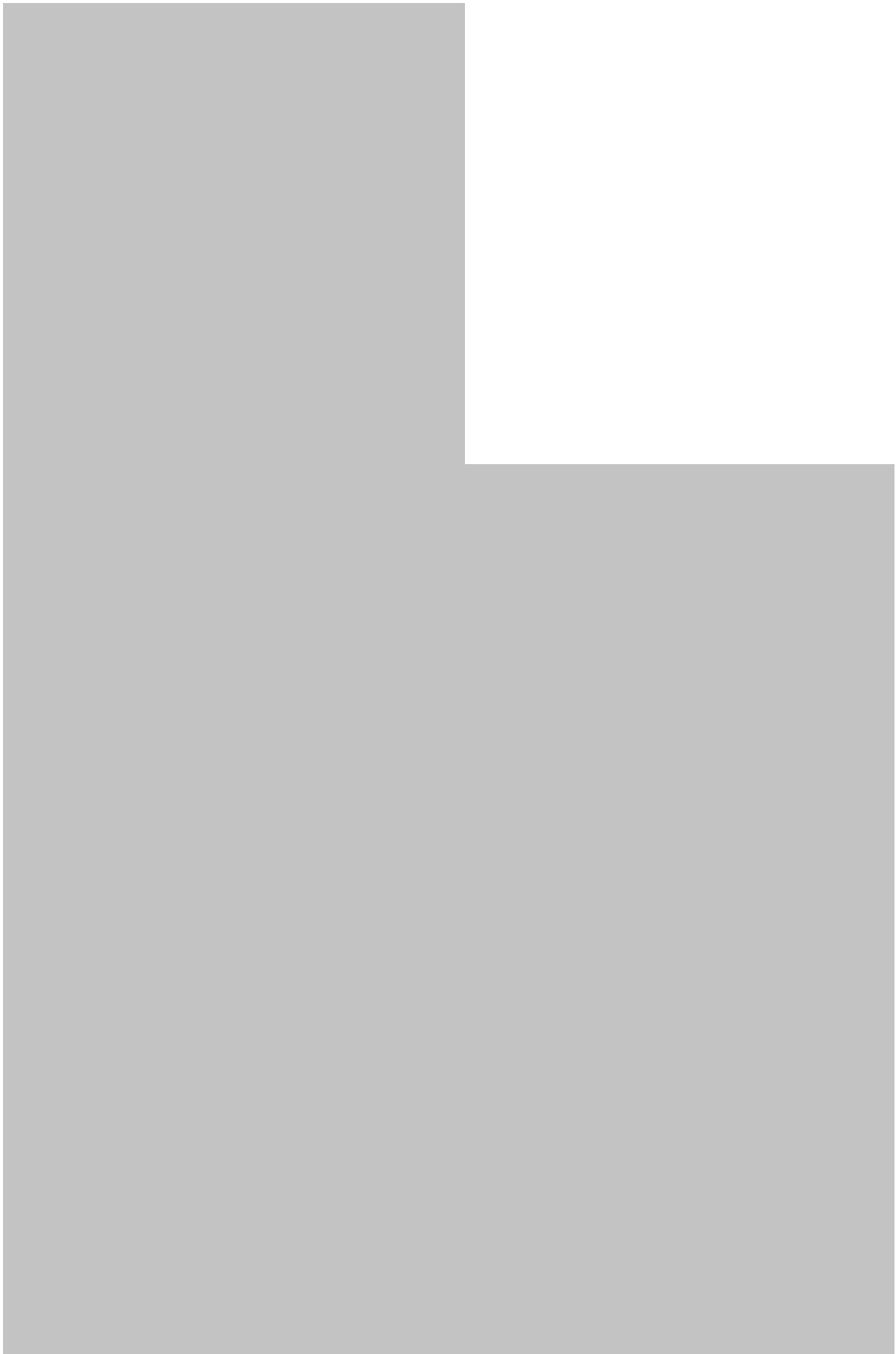
Comparison

Statistical methods are not interchangeable.

Analytical Goal	Useful Method	Typical Python Tool
Summarize data	Descriptive statistics	pandas
Measure spread	Variance/standard deviation	NumPy/pandas
Explore distribution	Histogram	Matplotlib
Detect outliers	Box plot	Seaborn
Compare averages	t-test	SciPy
Compare several groups	ANOVA	SciPy/Statsmodels
Measure relationship	Correlation	pandas/SciPy
Predict outcomes	Regression	Statsmodels/scikit-learn
Analyze categories	Chi-square	SciPy
Simulate probability	Random sampling	NumPy







Diagrams & Tables

A useful statistical-analysis architecture can be represented as:

Raw Data → Cleaning → Exploration → Descriptive Statistics → Visualization → Statistical Test → Interpretation → Decision

Another useful framework is:

Question → Variable Types → Distribution → Assumptions → Method → Result → Practical Meaning

Stage	Main Question
Collection	Where did the data come from?
Cleaning	Is the data reliable?
Exploration	What patterns exist?
Description	What does the dataset look like?
Inference	What can the sample tell us?
Modeling	Can relationships be quantified?
Interpretation	What does the result actually mean?

Exercises 51–60: Correlation

51. Calculate Pearson correlation

Solution: Use SciPy or pandas correlation functionality for numerical variables.

52. Detect positive association

Solution: A positive correlation indicates that variables tend to increase together.

53. Detect negative association

Solution: A negative correlation indicates that one variable tends to increase as the other decreases.

54. Find weak relationships

Solution: Inspect correlation values and visualize the variables with scatter plots.

55. Create a correlation matrix

Solution: Use pandas correlation functionality and visualize the result as a heatmap.

56. Find strongly related variables

Solution: Search the correlation matrix for relationships that are both statistically and practically meaningful.

57. Create a scatter plot

Solution: Plot one numerical variable against another to visually inspect their relationship.

58. Investigate nonlinear relationships

Solution: A low linear correlation does not necessarily mean that variables have no relationship; visualize the data.

59. Compare correlation groups

Solution: Calculate correlations separately for different populations or categories.

60. Avoid correlation mistakes

Solution: Remember that correlation alone does not establish causation.

Exercises 61–70: Hypothesis Testing

61. Formulate a research question

Solution: Convert the practical question into a testable statistical hypothesis.

62. Define the null hypothesis

Solution: State the default assumption that the analysis will evaluate against evidence.

63. Define the alternative hypothesis

Solution: Specify the effect or difference that the researcher wants to investigate.

64. Perform a t-test

Solution: Select the appropriate SciPy t-test according to whether groups are independent or paired.

65. Compare two independent groups

Solution: Use an independent-samples statistical test after checking assumptions.

66. Compare before-and-after measurements

Solution: Use a paired test when the same subjects or units are measured twice.

67. Interpret a p-value

Solution: Compare the p-value with the selected significance threshold and consider practical importance.

68. Analyze statistical significance

Solution: Statistical significance indicates evidence against the null hypothesis, not necessarily a large or useful effect.

69. Report test results

Solution: Include the test type, relevant statistic, p-value, and practical interpretation.

70. Avoid p-value-only decisions

Solution: Combine significance testing with effect size, confidence intervals, data quality, and domain knowledge.

Exercises 71–80: Regression

71. Create a simple regression model

Solution: Use Statsmodels or scikit-learn to model a numerical outcome using a predictor.

72. Examine a regression coefficient

Solution: Interpret the coefficient as the expected change in the response associated with a change in the predictor, within the model.

73. Analyze model fit

Solution: Examine appropriate goodness-of-fit measures and diagnostic plots.

74. Create predictions

Solution: Supply new predictor values to the trained model and generate predicted outcomes.

75. Plot regression results

Solution: Combine a scatter plot with the fitted regression relationship.

76. Analyze residuals

Solution: Examine residual patterns to identify nonlinearity, unequal variance, or unusual observations.

77. Add another predictor

Solution: Extend the model when another variable provides meaningful explanatory information.

78. Compare models

Solution: Evaluate predictive performance and statistical diagnostics rather than relying on model complexity.

79. Detect overfitting

Solution: Compare training and validation performance and avoid unnecessary variables.

80. Interpret regression responsibly

Solution: Regression can reveal associations and predictions but does not automatically prove causality.

Exercises 81–90: Advanced Analysis

81. Perform ANOVA

Solution: Use ANOVA when comparing the means of multiple groups.

82. Analyze categorical variables

Solution: Convert categories into suitable statistical representations and inspect their frequencies.

83. Perform chi-square analysis

Solution: Use SciPy's chi-square procedures to investigate relationships between categorical variables.

84. Create grouped summaries

Solution: Use pandas groupby to calculate statistics separately for different categories.

85. Calculate group averages

Solution: Group observations by category and calculate the desired numerical summary.

86. Compare group variability

Solution: Calculate dispersion statistics within each group.

87. Analyze time-dependent data

Solution: Convert dates to appropriate datetime formats and examine observations chronologically.

88. Detect trends

Solution: Visualize observations over time and investigate whether a systematic pattern exists.

89. Identify anomalous observations

Solution: Combine statistical rules with domain knowledge to identify potentially unusual records.

90. Build an analytical report

Solution: Combine tables, visualizations, statistical results, and written interpretation into a coherent report.

Exercises 91–100: Professional Data Analysis

91. Analyze customer ratings

Solution: Calculate distribution summaries, identify unusual ratings, and compare ratings between customer segments.

92. Analyze manufacturing measurements

Solution: Measure process consistency and investigate observations outside expected operating ranges.

93. Analyze employee performance

Solution: Compare performance distributions while controlling for relevant departments or job categories.

94. Analyze website traffic

Solution: Examine traffic distributions, trends, geographic segments, and relationships with engagement metrics.

95. Analyze sales data

Solution: Group sales by product, region, and time period to identify important patterns.

96. Analyze financial observations

Solution: Examine central tendency, volatility, distributions, and relationships between financial variables.

97. Analyze experimental data

Solution: Define hypotheses, inspect the data, select appropriate tests, and report uncertainty.

98. Create an automated statistical pipeline

Solution: Combine data loading, cleaning, analysis, visualization, and reporting into reusable Python functions.

99. Build a statistical dashboard

Solution: Present important indicators through charts, summary statistics, filters, and clearly labeled metrics.

100. Complete an end-to-end analysis

Solution: Start with a practical question, clean the dataset, explore distributions, calculate descriptive statistics, test relationships, build an appropriate model, validate the findings, and communicate the result.

Examples Without Equations

Example 1: Manufacturing Quality

An engineering company records measurements from manufactured components.

Python can identify the typical measurement, variation, extreme observations, and changes between production batches.

The analyst might discover that one production line has substantially greater variability than another.

The correct conclusion is not simply that one line has a different average. The analyst should investigate **why the variation differs**.

Example 2: Website Analytics

A website collects information about sessions, page views, devices, countries, and engagement.

Python can group the observations by country and device type. Statistical summaries may reveal that mobile users behave differently from desktop users.

The finding can support decisions about page design, performance optimization, and content strategy.

Example 3: Engineering Experiment

Suppose engineers compare two materials under similar laboratory conditions.

Python can summarize measurements, visualize their distributions, compare groups statistically, and identify unusual observations.

The final engineering decision should consider statistical evidence together with material cost, safety, durability, and manufacturing requirements.

Real-World Application

Statistics with Python has applications across almost every technical industry. 🌍

Engineering

Engineers use statistics for:

- Quality control
- Reliability analysis
- Experimental design
- Process optimization
- Sensor analysis
- Failure investigation

Data Science

Data scientists use statistical techniques for:

- Exploratory data analysis
- Feature analysis
- Model evaluation
- Experimentation
- Forecasting
- Anomaly detection

Business

Organizations use statistics to understand:

- Customer behavior
- Sales performance
- Market trends
- Operational efficiency
- Product performance

Healthcare and Research

Researchers use statistical analysis to evaluate experimental results, compare populations, investigate relationships, and quantify uncertainty.

Finance

Statistical techniques support:

- Risk analysis
- Portfolio research
- Market analysis
- Performance measurement
- Financial forecasting

Common Mistakes

Mistake 1: Ignoring Data Quality

Garbage data can produce misleading statistical conclusions.

Mistake 2: Using the Mean Automatically

The mean may be strongly affected by extreme observations.

Mistake 3: Confusing Correlation with Causation

Two variables can move together without one causing the other.

Mistake 4: Choosing Tests Before Understanding Data

The statistical method should follow the research question, variable types, study design, and assumptions.

Mistake 5: Ignoring Visualization

A single statistical summary can hide important distribution patterns.

Mistake 6: Treating Statistical Significance as Practical Importance

A statistically significant result may have little real-world impact.

Mistake 7: Forgetting Sampling Bias

A technically correct analysis can still produce poor conclusions when the sample does not represent the population.

Challenges & Solutions

Challenge	Solution
Missing values	Investigate why values are missing before selecting a treatment
Outliers	Combine statistical detection with domain knowledge
Small samples	Report uncertainty and avoid overconfident conclusions
Non-normal data	Consider robust or non-parametric methods
High-dimensional data	Use structured exploratory analysis and appropriate visualization
Confounding variables	Consider study design and multivariable analysis
Reproducibility	Keep analysis code, data-processing steps, and documentation together
Misinterpretation	Translate statistical outputs into practical language

Case Study

Manufacturing Process Investigation

Consider a hypothetical factory producing precision mechanical components.

The engineering team observes that customer complaints have increased. The production database contains measurements from several manufacturing machines.

The first stage is data inspection. Python reveals missing observations, duplicate records, and inconsistent machine labels.

After cleaning, the analyst summarizes measurements by machine.

A visualization shows that most machines produce relatively consistent measurements, while one machine produces a much wider distribution.

The team then investigates the machine's operating conditions. Additional analysis indicates that its measurements change considerably during longer production

periods.

Rather than changing the entire manufacturing process, engineers focus their investigation on that specific machine.

The statistical analysis therefore becomes a **decision-support tool** rather than merely a collection of numerical calculations.

This illustrates an important professional principle:

The goal of statistics is not to produce numbers; it is to produce reliable evidence for better decisions.

Essential Tips

For Beginners

-  Learn pandas before attempting advanced statistical modeling.
-  Always visualize numerical data.
-  Understand your variables before selecting a test.
-  Treat data cleaning as part of statistical analysis.
-  Write down the analytical question before writing Python code.

For Advanced Users

-  Check assumptions before applying statistical procedures.
-  Report uncertainty rather than presenting estimates without context.
-  Combine statistical evidence with domain expertise.
-  Distinguish exploratory analysis from confirmatory analysis.
-  Make analytical workflows reproducible.
-  Focus on effect size and practical significance, not only statistical significance.

FAQs

What is the best Python library for statistics?

There is no single best library for every task. pandas is excellent for data manipulation and descriptive analysis, SciPy provides many statistical procedures, and Statsmodels is particularly useful for statistical modeling and inference.

Is Python difficult for [learning statistics](#)?

Python can make statistics easier because repetitive calculations can be automated. However, you still need to understand statistical concepts, assumptions, and interpretation.

Can beginners use these 100 exercises?

Yes. The exercises progress from basic data inspection and descriptive statistics toward more advanced analytical methods.

Should I learn mathematics before statistics with Python?

A basic understanding of mathematical concepts is useful, but beginners can start with practical statistical reasoning and gradually develop the mathematical foundations.

What should I learn after descriptive statistics?

A logical progression is probability, distributions, sampling, hypothesis testing, correlation, regression, and experimental design.

Is correlation enough to prove a relationship?

No. Correlation measures association between variables. It does not by itself demonstrate that one variable causes another.

Why are visualizations important in statistics?

Visualizations reveal patterns, distributions, outliers, clusters, and trends that can be difficult to recognize from numerical summaries alone.

Can Python statistics be used in engineering?

Absolutely. Statistical Python workflows are useful for quality control, reliability studies, experiments, manufacturing analysis, sensor data, and process optimization.

Conclusion

Statistics with Python provides a powerful bridge between **raw data and evidence-based decision-making**. 📊🔗

The 100 exercises in this guide progress from basic data inspection through descriptive statistics, probability, distributions, correlation, hypothesis testing, regression, ANOVA, categorical analysis, and professional analytical workflows.

The most important lesson is that statistical analysis is more than calling a Python function. A strong analyst asks:

What question am I trying to answer?

Is the data suitable for answering it?

Which statistical method is appropriate?

What assumptions apply?

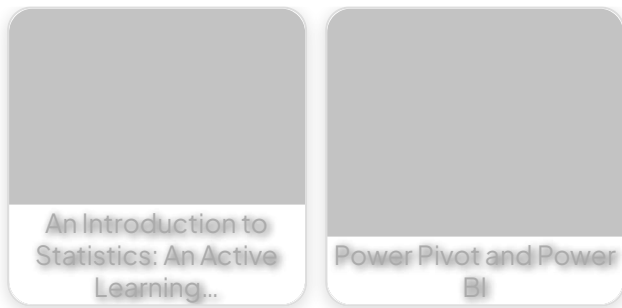
What does the result actually mean?

When Python programming is combined with sound statistical reasoning, students and professionals can analyze complex datasets efficiently while producing conclusions that are clearer, more reproducible, and more useful.

🚀 **Learn the statistics. Master the Python tools. Question the data. Then turn evidence into engineering and business decisions.**

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